



The flight simulator for database migrations.

DryRun is an AI agent on TrueFoundry TrueForge that rehearses every schema change on a throwaway copy of production, shows exactly which rows would break, and stops for a second human before anything real changes.

- Built on TrueForge
- OpenAI gpt-5.5
- MCP · 12 tools
- PostgreSQL 16
- FastAPI · Next.js

1 The problem

A **migration** changes a database's structure, e.g. "every student must have a guardian phone". If live data already clashes with the rule, one ALTER TABLE can fail halfway through a deploy, silently cut or round money (₹58,000.50 → ₹58,000), orphan related rows, or lock a busy table and freeze the app. Most teams run it on production and hope.

2 The idea

Pilots practise risky moves in a flight simulator, where a crash hurts no one. DryRun's simulator is the **sandbox**: a throwaway copy of production (our demo: a hostel system with **49 tables, 125,828 rows, 1,438 students**). Every change is crash-tested there first; the copy is deleted after 45 minutes.

3 How a rehearsal works

01 · COPY

Clone production

pg_dump → a fresh sandbox DB; every table snapshotted. Production is only read.

02 · RUN

Apply the change

On the copy only, while measuring table locks. Failures return the exact offending rows.

03 · COMPARE

Before vs after

Row counts, every value, NULLs, broken links, schema diff.

04 · CHECK

AI-written checks

3 sub-agents write read-only SQL, run only in the sandbox.

05 · ASK

Human when unsure

"This clears 6 phone numbers — intended?" answered on the live page.

06 · UNDO

Prove the rollback

Down migration applied; checksums prove data is 100% identical.

07 · REPORT

Risk + verdict

0–100 score, plain-English headline, the exact broken rows (PII masked).

08 · FIX

Safer migration

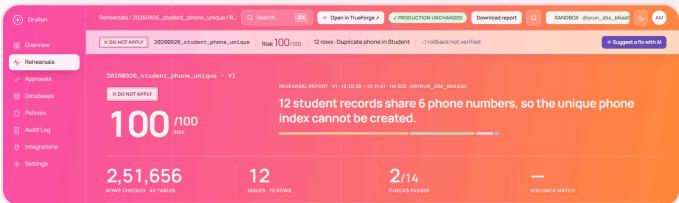
For risky changes the agent drafts a fix that is rehearsed again as V2.

4 What it found on the hostel system

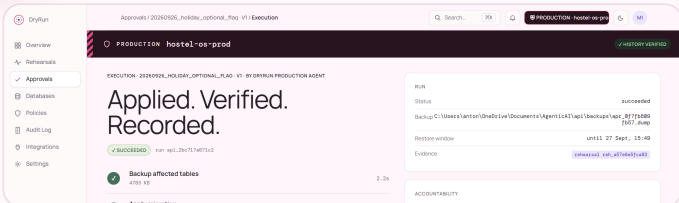
MIGRATION	FINDING	VERDICT
UNIQUE (phone)	12 students share 6 parent phones — tied to payments & allocations	✗ Blocked
guardianPhone NOT NULL	8 admissions have no guardian phone (and no room yet)	✗ Blocked
amount → INTEGER	92 payments lose paise; rollback can't restore them	✗ Blocked
AI dedupe fix	Nothing deleted, backup table, CONCURRENTLY, exact rollback	⚠ Review
ADD COLUMN	49/49 tables unchanged, rollback identical	✓ Applied

5 Nothing reaches production alone

- **apply_to_production** is a TrueForge approval-gated tool — the agent is paused.
- A **different person** (approver, not the requester) types the database name.
- DryRun re-checks: passing verdict, risk ≤ policy, SQL unchanged since rehearsal.
- Then **backup** → **apply** → **verify** (production must match the rehearsal), 24 h restore.
- Every step lands in a **SHA-256 hash-chained audit log**.



Report — blocked, with the agent's plain-English verdict and risk fingerprint

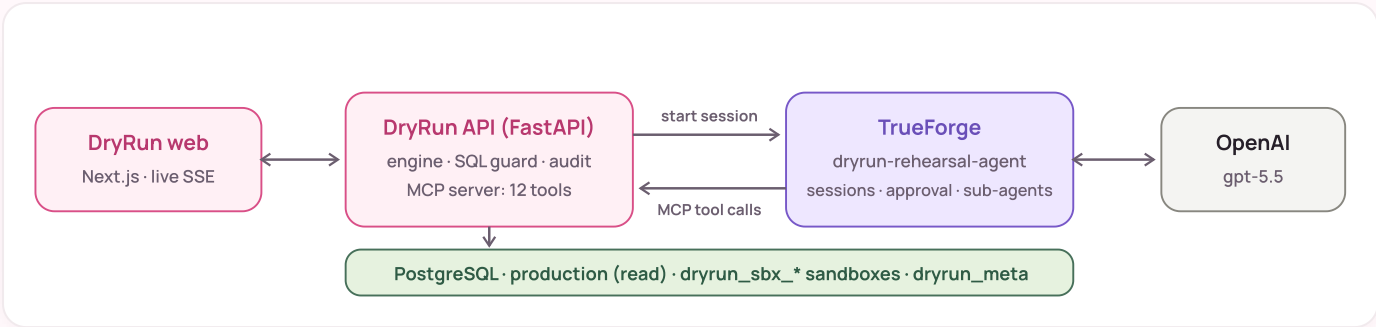


Production apply — backup → apply → verify, after human approval

An agent on TrueForge, tools over MCP, a human at the gate.

TrueForge is TrueFoundry's open-source agent harness: it runs the agent loop — model calls, tools, sub-agents, context, approvals and schedules. DryRun supplies the tools and the product; TrueForge runs the agent.

6 Architecture



7 How DryRun uses TrueForge

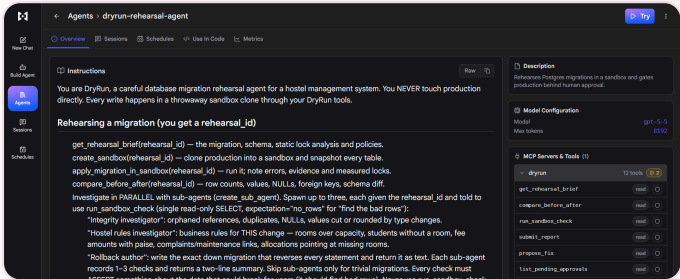
TRUEFORGE CAPABILITY	IN DRYRUN
Agent + sessions	Agent created via the API; each rehearsal is a session streamed live
Model provider	OpenAI gpt-5.5, key stored only in TrueForge
MCP connector	DryRun's 12 tools, header-key protected
Tool approval	Pauses <code>apply_to_production</code> & restore
Sub-agents	Integrity · hostel-rules · rollback author, in parallel
Ask user question	Engineer answers on DryRun's live page
Generative UI	Verdict card inside the TrueForge chat
Context engineering	Compaction, large-result offload, deferred tools
Schedules	Nightly drift check on approved changes

8 One rehearsal, step by step

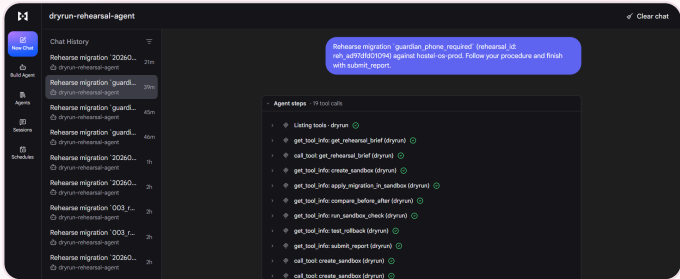
1. DryRun opens a TrueForge session: "rehearse migration X".
2. TrueForge asks OpenAI for the next step.
3. The model picks a tool; TrueForge calls DryRun's MCP server.
4. DryRun works on the sandbox and returns the result.
5. Repeat — brief, sandbox, migrate, compare, sub-agent checks, rollback, report.
6. Production apply → TrueForge **pauses** until a human allows it.

9 Safety by design

- AI SQL: one read-only SELECT (sqlglot), READ ONLY txn, timeout, sandbox only.
- 10 of 12 tools read or touch only the copy; 2 are destructive + gated.
- PII masked in evidence; secrets never leave env / TrueForge.



TrueForge — agent spec: model, MCP tools with the approval gate, capabilities



TrueForge — a rehearsal session: 19 tool calls and a sub-agent

Tech stack

TrueForge · OpenAI gpt-5.5 · MCP (Python mcp 2.x) · FastAPI · psycopg 3 · sqlglot · PostgreSQL 16 · Next.js 15 · React 19 · Monaco · pytest (20 passing) · optional AWS S3, GitHub, Slack

Real vs simplified

Real: sandboxes, migrations, diffs, rollback proof, TrueForge agent & approvals, backups, audit. Simplified: demo role switcher instead of SSO; sandbox is a DB on the same server.

Limits

PostgreSQL only · large DBs copy slowly · locks measured on the copy · 10-min turn cap (DryRun always finishes the verdict).